The MMH Survey

Surveys are tools that can remind us of conditions and practices that, when present in the workplace, can lead to injury or illness. Surveys should ask critical questions about the operation. The answers, if the questions are properly designed, will lead the inspector to general conclusions with regards to the degree of risk. A Manual Materials Handling survey should include, at the minimum, questions similar to the following:

Costs

	Yes	No
Are we experiencing costly accidents relative to this task?	☐	☐

Activities

Are any of the following activities required in this task?

Lifting / Lowering	☐	☐
Pushing / Pulling	☐	☐
Twisting	☐	☐
Carrying	☐	☐
Holding	☐	☐

Models

Do any of the following models apply to this task?

Biomechanical (compression forces at L5)	☐	☐
Physiological (frequency vs. weight)	☐	☐
Aerobic Capacity (conditioning)	☐	☐
Effective Use of Muscles	☐	☐
NIOSH Lifting Guidelines	☐	☐

Factors

Are any of the following influencing the task?

Worker Characteristics	☐	☐
Task Characteristics	☐	☐
Material / Container Characteristics	☐	☐

Training

Is the employee in need of training in safe lifting guidelines?	☐	☐
Is the employee in need of training in body mechanics?	☐	☐

Note: Any one "Yes" answer can be serious enough to cause serious injury or illness. The more "Yes" answers the higher the risk even when the single condition is determined to be minor. This is due to the cumulative effect of multiple negative factors.

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Safe Lifting Guidelines

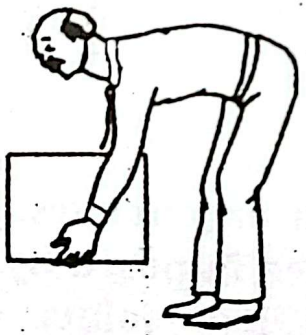
The following five general rules should be applied at every opportunity.

- Plan the lift.
- Both squat and stoop lifting is now considered acceptable for jobs requiring repetitive lifting. The term used to describe this is free form lifting. No matter wha type of lift is used, it is never permissable to exceed the maximum acceptable load of the worker.
- Keep the load as close to the body as possible.
- Lift the load with a smooth body motion. *(Avoid jerking)*
- When turning, do not twist. Turn with the feet rather than twisting of the trunk.

One your own: Describe what, if anything, you see in each of the following pictures.

Worker picking up box: _____

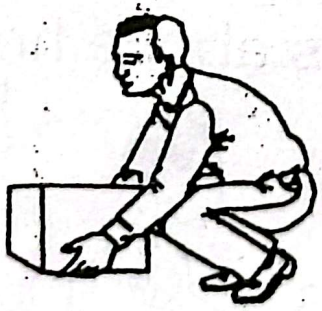
What would you want to see done differently?



(Adapted from: The Saunders Group inc., 4250 Norex Drive, Chaska, MN 55318)

Worker picking up box: _____

What would you want to see done differently?



(Adapted from: The Saunders Group inc., 4250 Norex Drive, Chaska, MN 55318)

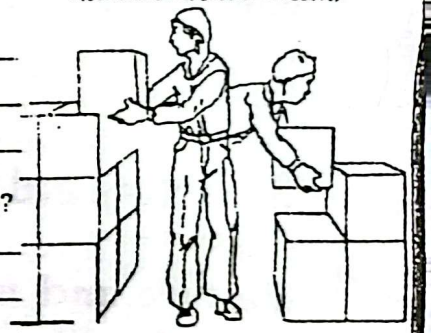
Worker lifting patient from wheelchair:

What would you want to see done differently?

(Adapted from: The Saunders Group inc., 4250 Norex Drive, Chaska, MN 55318)

Worker lifting and moving boxes:

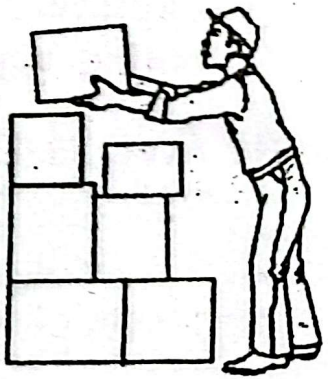
What would you want to see done differently?



(Adapted from: The Saunders Group inc., 4250 Norex Drive, Chaska, MN 55318)

Worker stacking boxes:

What would you want to see done differently?



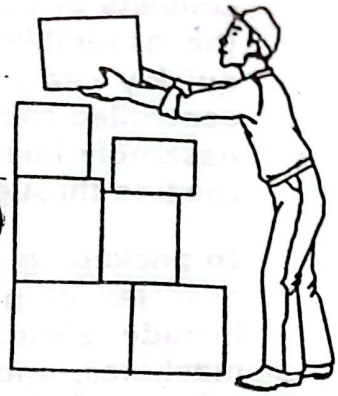
(Adapted from: The Saunders Group inc., 4250 Norex Drive, Chaska, MN 55318)

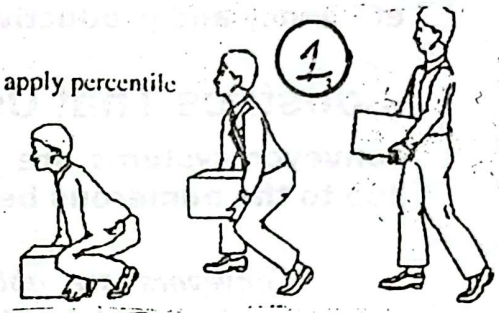
EVIEW QUESTIONS

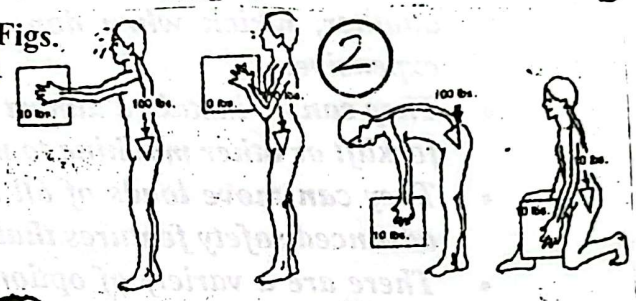
1. What does the word *ergonomics* mean?
2. Describe what the field of ergonomics is about.
3. How is ergonomics applied?
4. Identify three things that people are good at and three that machines are good at.
5. Name two fundamental principles of ergonomics.
6. What is anthropometry?
7. Much anthropometric data are given in percentiles. Describe how to apply percentile data in design.
8. Name three principles for applying anthropometry in design.
9. Describe the six elements in the people—machine interface.
10. What is biomechanics? Kinesiology? Kinematics? Kinetics?

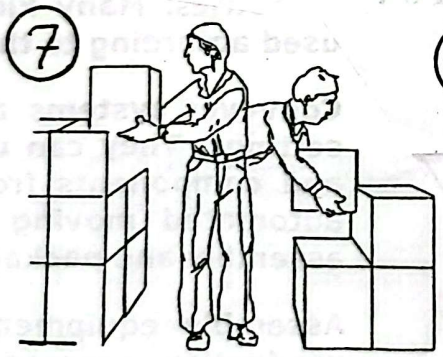
11 - What is anthropometric?

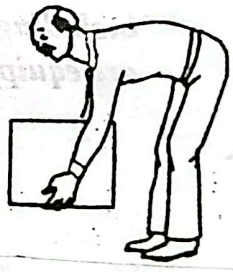
12 - Select Correct and Incorrect Techniques Figs. 1 To 12

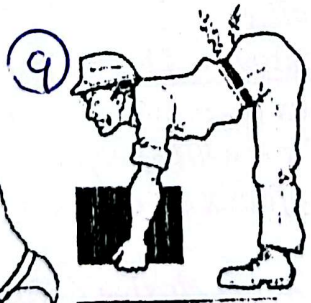


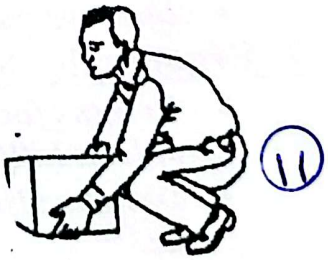












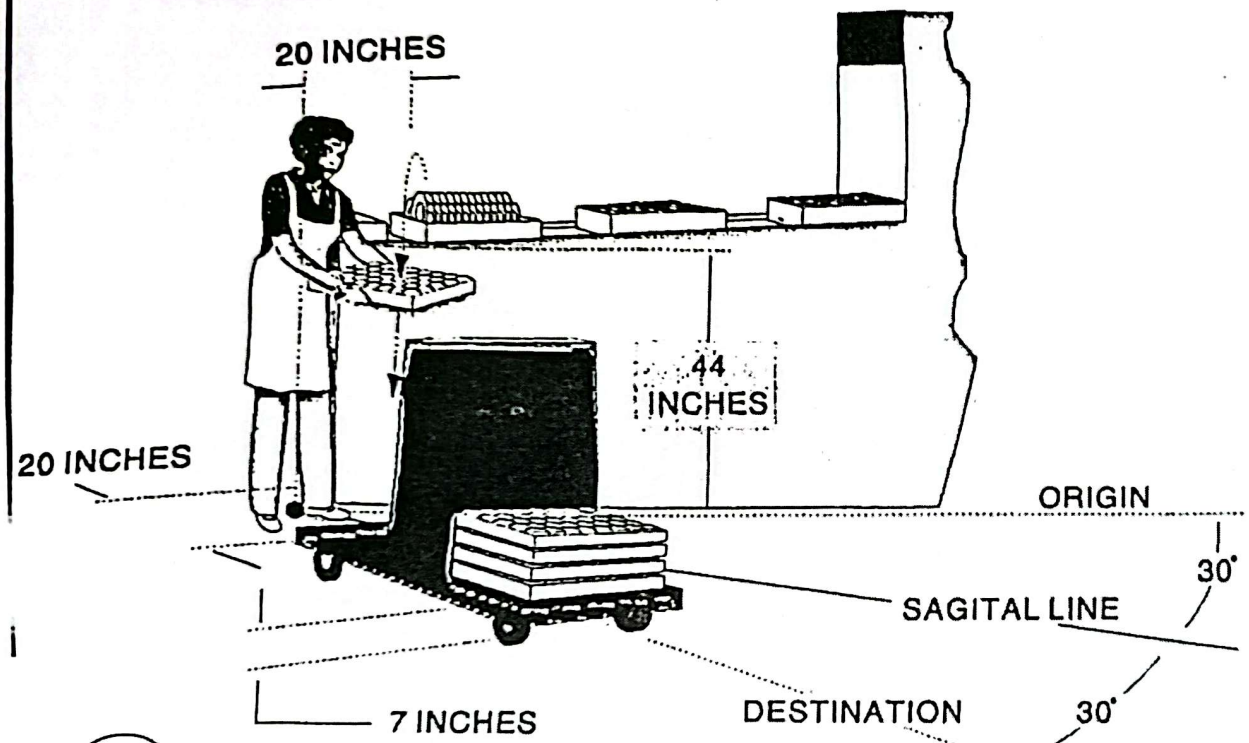


Figure 15 Dish-Washing Machine Unloading, Example 5

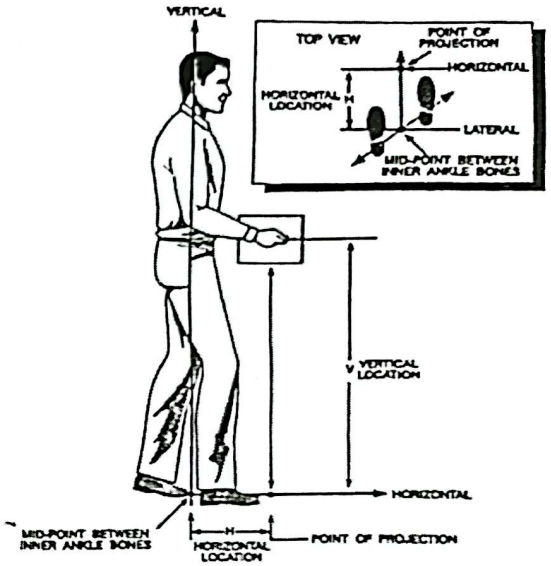


Figure 1 Graphic Representation of Hand Location

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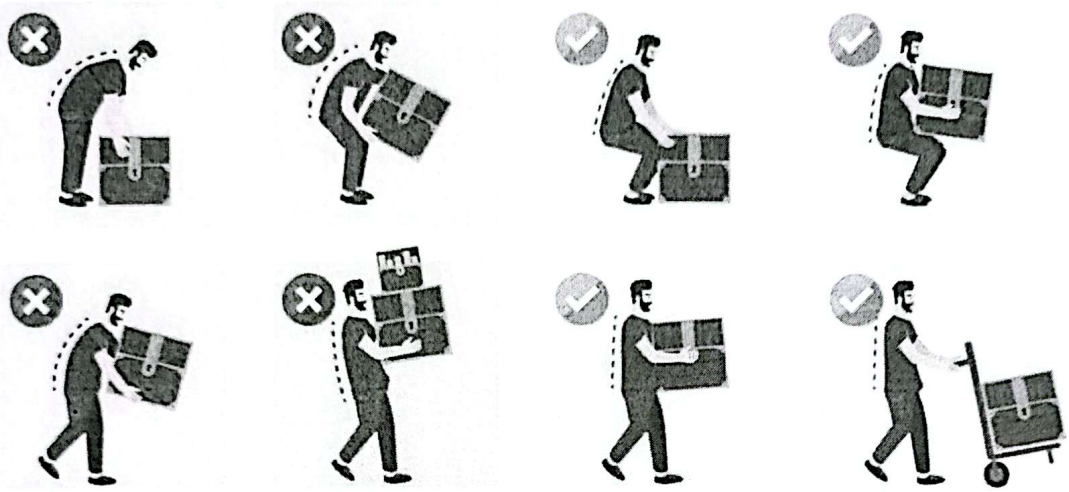
3. **Excessive Force**: Tasks that require lifting, pushing, or pulling heavy loads can overexert muscles and joints.



4. **Static Postures**: Maintaining the same position for prolonged periods can reduce blood flow and cause muscle fatigue.

5. **Vibration**: Regular exposure to vibrations from tools or machinery can lead to nerve damage and circulatory problems.

Q.7- How Do Hand Tools Cause Back Pain?

ANS **Some types of industrial work involve the use of hand and power tools. At first you may think that hand and power tool use only involves the fingers, hands, and arms. This is not always true. Tool use often demands total body involvement. For example, tool use may require standing or sitting in one position (static posture) for extended time periods, repetitious movements, or excessive force. Stationary postures or excessive movement of any type can stress the spine and, depending on the body action needed to use the tool, can cause neck and back problems.**
There are hundreds of different types of tools. Tools vary by shape, size, weight, and power source. Some tools are simple to use and others are complicated to operate. A side effect from the use of certain power tools is vibration, which can affect the entire body and cause neck and back pain, numbness, tingling, aching, stiffness, headache, or blurred vision.

Q.8- What is the NIOSH Lifting Equation?

ANS [NIOSH *National Institute for Occupational Safety and Health*]

The NIOSH Lifting Equation is a tool used to identify, evaluate, or classify some risks associated with a lifting task.

NIOSH **updated this tool by issuing the** ***Revised 1991 NIOSH Lifting Equation.***
The new equation increased the types of task variables the tool could assess.

Q.9- What are the Lifting Techniques?

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5)

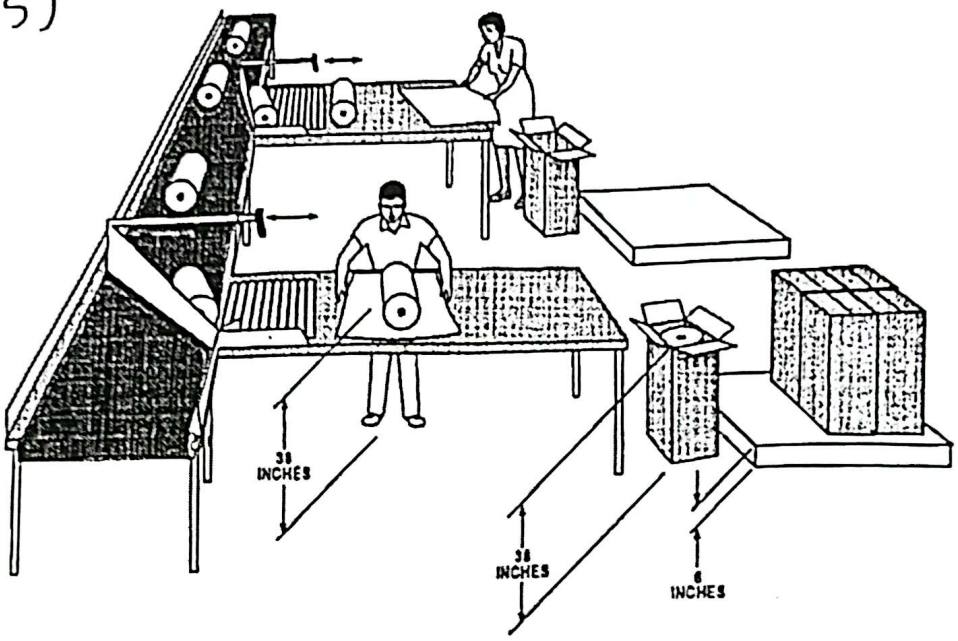

Figure 25 Product Packaging II, Example 9

6)

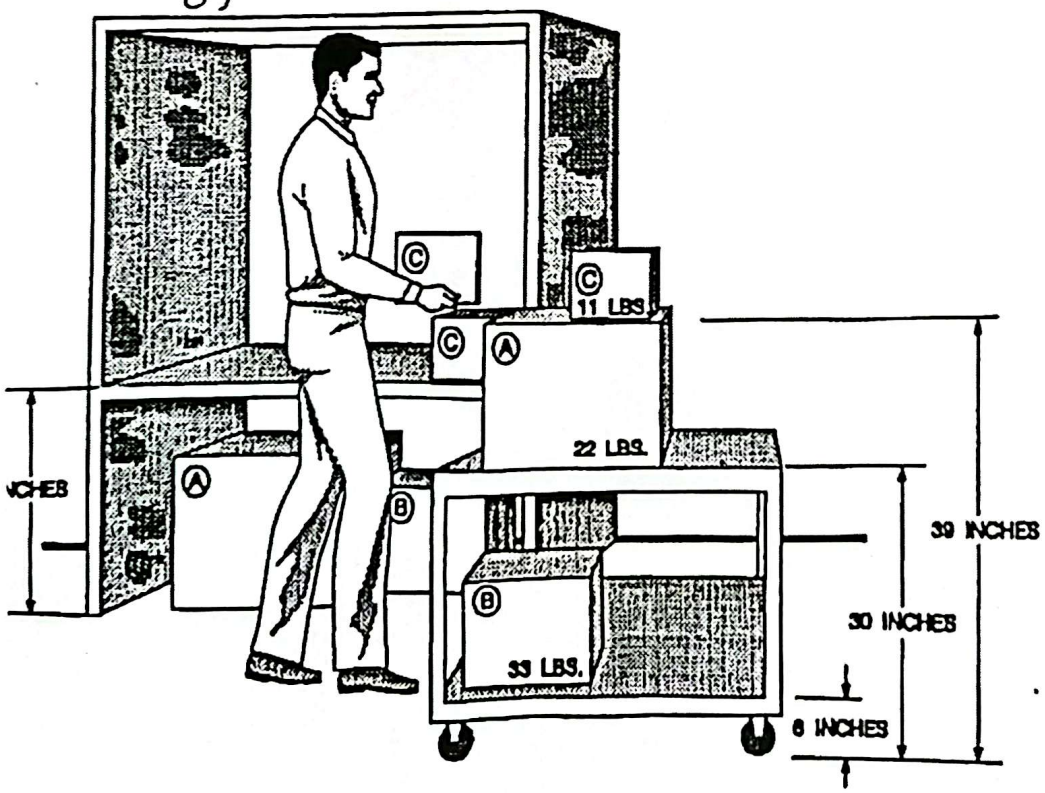

Figure 27 Warehouse Order Filling, Example 10

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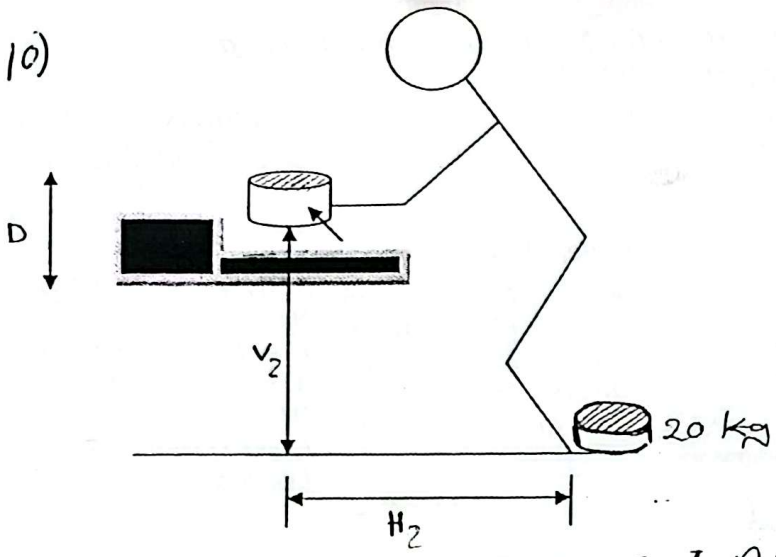
10)



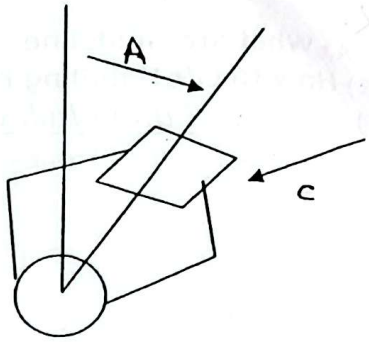
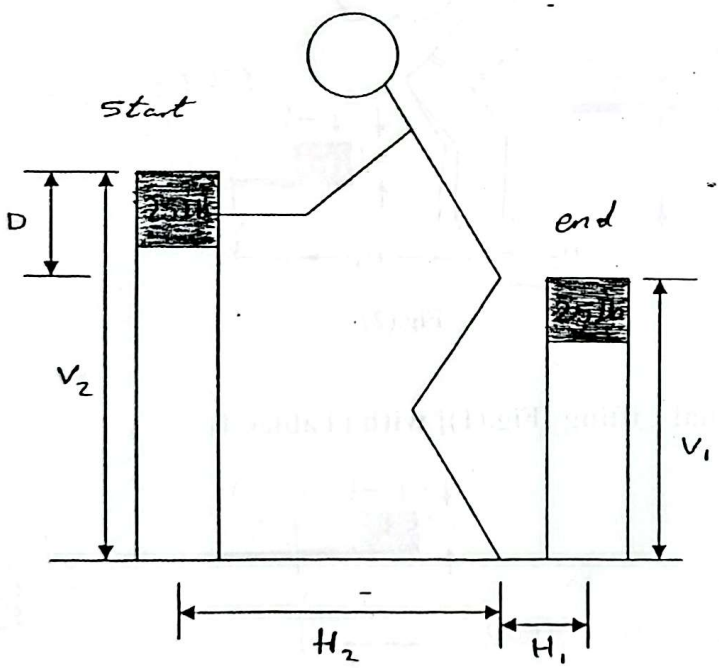


Fig. (1) Find RWL by Any Methods

11)



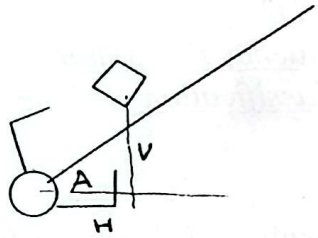


Fig. (2)
Find Lifting Index